

Milestone 1 Project Plan

COMPLETION RECORD · ACCEPTANCE EVIDENCE · NEXT-MILESTONE GATE

Milestone outcome

Milestone 1 is complete at semantic version 0.1.0. It delivers the hull only and does not start any later content milestone.

Completed work packages

- Requirements and public-reference boundary — complete
- Parameterized sixteen-station full-hull loft — complete
- Bow/stern/waterline/anchor geometry — complete
- Automatic A1 Mini-compatible splitting — complete
- Four-shaft running gear and twin rudders — complete
- Hidden self-aligning glue joints — complete
- FCStd / STEP / STL / 3MF / OBJ exports — complete
- Geometry QA, renders, drawings, BOM, and guides — complete

Acceptance record

Acceptance item	Result	Evidence
Complete kit part count	PASS	detected 21 disconnected watertight parts
Preferred plate envelope	PASS	Hull.stl bounds [220.0562, 177.29082, 31.43096] mm
A1 Mini module envelope	PASS	module maximum axes [164.66667, 164.76666, 158.76666] mm
3MF package and indices	PASS	74446 triangles; 4 allowed materials; CRC=True
FreeCAD BRep validity / self-intersection	PASS	23 leaf/reference BRep shapes passed OCC BOPCheck
STEP round-trip	PASS	21 closed solids after STEP re-import; 0 self-intersection issues (92 OCC p-curve diagnostics recorded)
Overall hull length	PASS	measured 476.000 mm; target 476.000 mm
Maximum molded hull beam	PASS	measured 58.304 mm; target 58.300 mm
Glue-joint fit allowance	PASS	0.250 mm per side; 0.50 mm diametral

Configuration

The 1:700 master is 476.0 mm long. The generator accepts 1:1000, 1:700, and 1:350 and derives split count from the scaled length. Functional gauges are clamped for a 0.4 mm nozzle.

Retained risks

- Physical first-article and adhesive fit remain required.
- Public shipyard body plans were unavailable; station geometry is a documented public-data reconstruction.
- STEP round-trip records OpenCascade p-curve diagnostics on analytic loft faces; all solids remain valid/closed with no self-intersection issue.
- Appendage gauges prioritize reliable PLA printing over exact scale thinness.

Milestone gate

v0.2 Flight Deck and all later milestones remain pending. No flight-deck detail, island, weapons, aircraft, radar, or display-base work is present in v0.1.0.

Public technical references

NAVSEA Cost Estimating Handbook — Nimitz-class overview

NAVSEA Naval Nuclear Propulsion Program — class dimensions

Naval History and Heritage Command — attack carriers